

Profile

Real-time 3D and software engineering leader with 8+ years across Unreal Engine, Unity, spatial visualization, custom 3D pipelines, and production systems. Architected a real-estate visualization platform spanning eight property models, improving asset load times by 45% through streaming and LOD optimization. Combines hands-on C++/C#/Python engineering with backend architecture, low-latency systems, and cross-functional leadership. Experienced in translating client requirements into technical solutions, leading multidisciplinary teams, and delivering interactive environments under live production constraints. Seeking to apply this background to large-scale Digital Twin and architectural visualization programs.

Core expertise	Real-time 3D	Unreal Engine 4/5, Unity, C++, C#, Blueprints, URP, WebGL, VR, interactive simulation
	Spatial & rendering	Architectural and real-estate visualization, model-to-runtime workflows, asset streaming, LOD optimization, large-scene performance profiling, spatial queries
	Data & integration	Python, C++ data pipelines, REST APIs, backend and service architecture, real-time state synchronization, cloud/edge deployment, data formats and throughput
	Technical leadership	Solution architecture, technical direction, cross-functional delivery, stakeholder consulting, engineering standards, sprint planning, hiring and mentoring
	Security & automation	Secure key and secret handling, auditable transaction flows, security-sensitive development, CI workflows, pipeline automation

- Experience**
- Software Engineer, Sensing & Data Systems** Mar 2026 – Present

DR3 · UAE

 - Architected a Wi-Fi CSI sensing platform for security and emergency-services applications, defining a three-layer backend architecture spanning sensing, digital signal processing, and decision output for embedded-edge and cloud deployment.
 - Built the Python signal-processing pipeline for noise filtering, drift removal, sub-carrier selection, and rhythmic-pattern detection, taking the system from concept to a working proof of concept.
 - Translated operational stakeholder requirements into pilot applications for privacy-preserving presence, motion-intensity, and vital-sign-band detection without cameras or wearables.
- Lead Software Engineer / Product Lead** Jul 2024 – Jul 2026

City Crafters · Dubai, UAE

 - Architected Hologram Cloud, an interactive real-estate visualization platform spanning eight property models. Reduced asset load times by 45% through streaming and LOD optimization; achieved +60 NPS in field tests.
 - Owned end-to-end technical solution design across real-time 3D/VR, mobile, and web products, translating client and business requirements into architectures, prototypes, production plans, and delivery priorities for 10+ branded products.
 - Built and scaled the founding cross-functional team from 0 to 12 across engineering, design, QA, and UI. Introduced agile sprint workflows that reduced feature-delivery time by 35%.

- Led Pit Protocol, a distributed VR pit-stop simulator for Yas Marina / Abu Dhabi GP activations with Emaar and Address Hospitality. Delivered a demo-ready prototype in five months with six synchronized stations, sub-20 ms state sync, and 98% uptime across live event days.
- Coordinated technical delivery and client-facing demonstrations, aligning engineers, artists, designers, and operational stakeholders around integration requirements, performance targets, and production readiness.

Software Engineer, Fintech Wallet

Feb 2025 – Oct 2025

Gaya Wallet (Blockmob Labs) · Remote

- Built production features for a secure, non-custodial multichain wallet in React Native across iOS and Android (100K+ installs), with auditable transaction flows, defensive key/secret handling, and strict review in a security-sensitive codebase.
- Implemented payment and chain integrations, including fiat on-ramp flows, multichain balance aggregation, swap-routing UX, and secure on-device key management.

C++ Software Engineer, 3D Pipeline

Sep 2023 – Dec 2023

Swave Photonics · Silicon Valley, CA

- Built a C++ 3D pipeline bridging Unity render output with proprietary holographic display hardware, enabling live interactive visualization on a specialized display platform.
- Collaborated with the algorithms team on data formats, throughput, and rendering integration; presented a working holographic demonstration at CES.

Senior Game Developer / Lead Game Designer

Aug 2021 – Aug 2023

Nirah Studios AB · Stockholm, Sweden

- Engineered Unreal Engine C++ runtime systems, including autonomous AI agents using Behavior Trees and Environment Query System (EQS), terrain navigation, spatial environment queries, and interaction systems.
- Developed reusable inventory, UI, and traversal systems; coordinated internal and external teams, reviewed implementation priorities, and managed an average \$250K quarterly budget.
- Led technical prototyping and cross-disciplinary delivery; the prototype helped secure \$600K+ in investment.

Earlier roles	Software Engineer / Game Designer Feb 2024 – Jul 2024 MasterMind GmbH · Germany Led two developers and three artists through alpha development of a story-driven interactive title, coordinating systems architecture, content pipelines, and delivery.
	Game Developer May 2018 – Aug 2021 Hyphonics Inc / X on Board · Toronto & Beirut Built Unity/C# educational and mobile products end-to-end, including UI, content systems, level spawning, and interactive application logic; demos drove national TV coverage and \$31K+ in grants and investment.

Education	B.S. in Computer Science 2024 American University of Beirut
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